

Reese Ludwick

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SUMMARY

AI engineer shipping LLM and agentic systems to production end-to-end — multi-agent LangGraph systems, RAG pipelines, and custom MCP servers for enterprise clients at Cognizant, plus autoclip.studio, a live SaaS built and operated solo. Owns the full slice — agent orchestration, FastAPI + Python backends, PostgreSQL/pgvector/Redis, React/TypeScript — turning fast-moving LLM tooling into dependable software for real users.

EXPERIENCE

Software Associate

Cognizant Technology Solutions

Remote

Dec 2025 – Present

- Develop and maintain GenAI and agentic AI systems (LangChain, LangGraph, Python) across R&D and client-facing deliverables.
- Design and implement RAG pipelines and MCP server integrations connecting enterprise tools to LLMs in production.
- Scope, prototype, and deliver AI-powered data-analysis tools with cross-functional teams.

Freelance AI Engineer & Full-Stack Developer

Independent Contract, WorldCovers / APMC

Remote

May 2026 – Present

- Paid contract engineer maintaining production web properties (chinaoverprints.com, ianthom.org) and building WorldCovers, a philatelic cataloging platform sponsored by the U.S. Philatelic Classics Society.
- Deliver features and automation on a 14-table PostgreSQL schema, Django REST backend, and React/TypeScript frontend.

GenAI Developer Contractor

Cognizant Technology Solutions

Remote

Jun 2025 – Dec 2025

- Built custom FastMCP servers and deployed external MCP integrations alongside AWS infrastructure to give agents dynamic tool access in production-style agentic workflows.
- Built and delivered GenAI proof-of-concept solutions with LangChain, LangGraph, and Python; applied prompt engineering and model evaluation to optimize LLM outputs.

IT Student Helper

James Rumsey Technical Institute

Hedgesville, WV

Aug 2019 – May 2021

- Resolved 300+ support tickets; helped a campus Wi-Fi overhaul cut latency 25%; set up 100+ AV/network systems.

PROJECTS

CreatorClip / autoclip.studio — Python, FastAPI, Celery, Deepgram, pgvector, Voyage AI, Anthropic API

- Live, multi-tenant production SaaS (autoclip.studio) with Stripe billing and a React/TypeScript frontend — turns long-form YouTube content into short-form clips, learning a creator's channel style via a custom "DNA" module.
- Async Celery pipeline: Deepgram transcription (WhisperX/AssemblyAI selectable) → Voyage AI + pgvector embeddings → ffmpeg rendering, with full YouTube OAuth publishing.

CFO Agent — Python, FastAPI, LangGraph, PostgreSQL, Redis, Anthropic API

- Personal-finance AI agent (in active development) on a FastAPI + LangGraph backend with PostgreSQL and Redis.
- Hybrid context architecture over a Fernet-encrypted vault with CSV/OFX import; built for multi-tenant productization.

Agentic Software Development Team — Python, LangChain, LangGraph, Amazon Bedrock (Cognizant R&D)

- Multi-agent LangGraph system with specialized agents for code generation, testing, security review, and supervision — automating the full software build lifecycle end to end.
- Integrated Amazon Bedrock and open-source ML models as swappable inference backends.

Customer Call Data Analysis & Insights Engine — Python, NLP, Sentence Transformers (Cognizant R&D)

- NLP clustering with sentence-transformer embeddings grouping large volumes of customer call data into semantic issue segments; agentic analysis feeding executive reporting.

TECHNICAL SKILLS

Languages: Python, SQL

AI / Agentic: LangChain, LangGraph, RAG pipelines, MCP servers (FastMCP), Amazon Bedrock, Anthropic API, Sentence Transformers, TensorFlow, pgvector, prompt engineering, model evaluation

Backend / Infra: FastAPI, Celery, Django REST, PostgreSQL, Redis, Docker

Frontend / Tooling / Cloud: React / TypeScript, AWS, Azure Fundamentals, Linux/Ubuntu, Git/GitHub, VS Code, Claude Code, Jupyter

EDUCATION

Bachelor of Science, Computer Science — Shepherd University (Aug 2021 – May 2025), Major GPA 3.43

CERTIFICATIONS

TensorFlow Developer Specialization (DeepLearning.AI) · Azure Fundamentals (AZ-900) · Linux + Git (Boot.dev) · GitHub Copilot & GenAI Context Engineering (Cognizant)